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4.1 Introduction

Hydrogen is the simplest atom which contains one electron and one proton. In contrast to other elements (except helium) its valence electron is directly in the sphere of action of the nucleus. It is invariably present in most of the compounds we come across in our daily life such as water, carbohydrate, proteins etc. As it has an unpaired electron, it is reactive and exists as a diatomic molecule (H_2). However, the abundance of hydrogen gas in the earth's atmosphere is very small.

4.1.1 Position in Periodic Table

The hydrogen has the electronic configuration of $1s^1$ which resembles with ns^1 general valence shell configuration of alkali metals and shows similarity with them as follows:

1. It forms unipositive ion (H^+) like alkali metals (Na^+, K^+, Cs^+)
2. It forms halides (HX), oxides (H_2O), peroxides (H_2O_2) and sulphides (H_2S) like alkali metals (NaX, Na_2O, Na_2O_2, Na_2S)
3. It also acts as a reducing agent.

However, unlike alkali metals which have ionization energy ranging from 377 to 520 kJ mol^{-1}, the hydrogen has 1,314 kJ mol^{-1} which is much higher than alkali metals.

Like the formation of halides (X^-) from halogens, hydrogen also has a tendency to gain one electron to form hydride ion (H^-) whose electronic configuration is similar to the noble gas, helium. However, the electron affinity of hydrogen is much less than that of halogen atoms. Hence, the tendency of hydrogen to form hydride ion is low compared to that of halogens to form the halide ions as evident from the following reactions:

$$\tfrac{1}{2} H_2 + e^- \rightarrow H^- \quad \Delta H = +36 \text{ kcal mol}^{-1}$$

$$\tfrac{1}{2} Br_2 + e^- \rightarrow Br^- \quad \Delta H = -55 \text{ kcal mol}^{-1}$$

Since, hydrogen has similarities with alkali metals as well as the halogens; it is difficult to find the right position in the periodic table. However, in most of its compounds hydrogen exists in +1 oxidation state. Therefore, it is reasonable to place the hydrogen in group 1 along with alkali metals as shown in the latest periodic table published by IUPAC.

4.1.2 Isotopes of Hydrogen

Hydrogen has three naturally occurring isotopes, viz., protium ($_1H^1$ or H), deuterium ($_1H^2$ or D) and tritium ($_1H^3$ or T). Protium ($_1H^1$) is the predominant form (99.985 %) and it is the only isotope that does not contain a neutron.

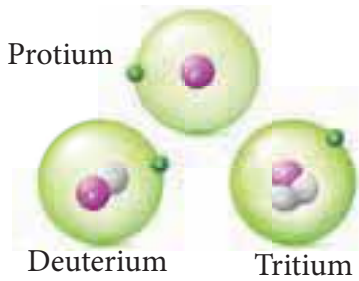


Figure 4.1 Isotopes of hydrogen

Deuterium, also known as heavy hydrogen, constitutes about 0.015 %. The third isotope, tritium is a radioactive isotope of hydrogen which occurs only in traces (~1 atom per 10^{18} hydrogen atoms). Due to the existence of these isotopes naturally occurring hydrogen exists as H_2, HD, D_2, HT, T_2, and DT. The properties of these isotopes are shown in Table 4.1.